2026/2027

ANALYSIS REPORT - ATAM

CLT ALT ELITE

Version: 1.0
Prepared by: CLT ALT ELITE Business Analyst Team

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AI support was also used to generate initial ideas, which were thoroughly reviewed, evaluated, and refined by the team before inclusion.

All substantive analysis, interpretation, and conclusions presented in this assignment remain our own work.

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1. Executive Summary

1.1 Background Information

This report was prepared by CTRL ALT ELITE Business Analysts (see Figure 1) on 7th February 2026 using the Project Specifications (PS) issued by our customer, Singapore University of Waterloo (SUW), dated 29th January 2026. The report summarises our team's findings with respect to the PS, which includes the assumptions made, identified incongruities and assessed requirement feasibility, as well as the timeline, resolutions and engagements with the client.

The investigative analysis method used is the "Breakdown, Clarify, Interpret & Categorise" (BCIC). The four BCIC activities were carried out sequentially under the regulated-revision framework as documented in section 2.

1.2 Project Context

Our client, SUW, had sought out our services to develop a timetabling and attendance management app. They have been facing difficulties with manually tracking staff and students using conventional paper-based methods for maintaining data, which is expensive and prone to irregularities. As SUW is a large institution, it could not afford to let such inconsistencies hamper its day-to-day operations. Moreover, they had noted that it has become increasingly difficult to effectively schedule lectures and maintain attendance records by merely relying on manual methods of data logging.

Therefore, the main objective of the system is to automate the scheduling of lectures and track students' attendance. The system also assists in classifying staff and students' records according to their respective subjects, thereby making data retrieval easy.

1.3 Organisational Chart

This analysis report is prepared under the direction of Project Lead Kevin Anthony Jones, with contributions from the business analysts across finance, software integration, frontend, backend, and external systems. The account manager supports stakeholder communication. The chart below illustrates the team's structure and reporting relationships.

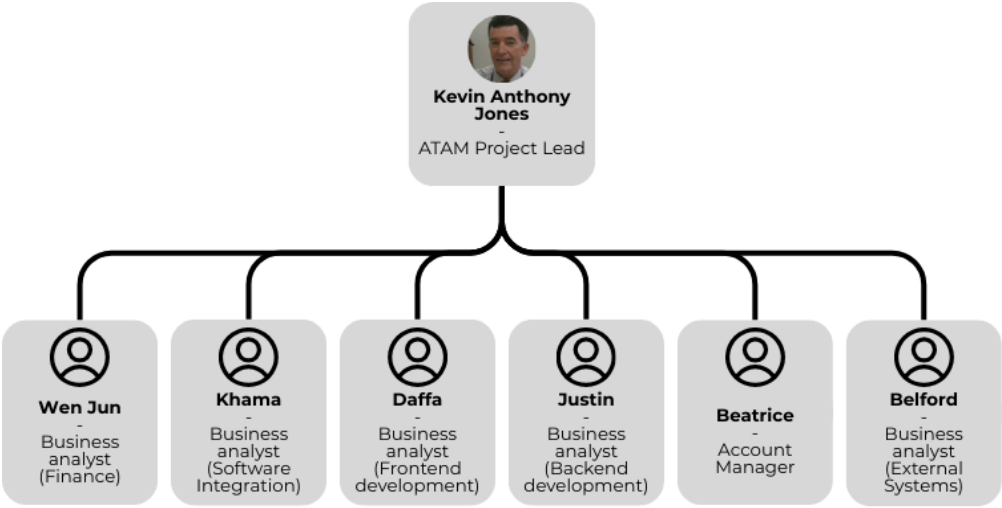

Figure 1: Organisational Chart For Project ATAM

2. BCIC Analysis

2.1. Breakdown

The breakdown of the Biz-Ops Requirements was marked against CTRL ALT ELITE's personal Incongruous Requirements Checklist (IRC), which can be found in the repository [Link to GitHub].

The IRC was applied during the Breakdown stage as a structured tool to classify problematic requirements. It evaluates requirements across four dimensions: Priority Keywords (to check the strength and correctness of imperative phrasing), Continuance (to identify complexity introduced by linking phrases), IRC Category (to classify issues such as ambiguity, inconsistency, or unrealistic statements), and Check Type (to specify the nature of the review, such as contextual, imperative, or testability). Each requirement flagged is annotated with its ID (as from the Customer's PS at this stage) and reviewed under the appropriate check type. The table below presents the team's IRC as applied to the ATAM PS, with each check linked to the specific PS clause it flagged.

Priority Keywords	**Continuance**	**IRC Category**	**Check Type**
Shall (strongest)	Below	Inconsistent with another requirement	Contextual Review
Must / Must not	As follows	Not Testable	Imperative Review
Is required to	Following	Unseemly Prioritized	Continuance Review
Are applicable	Listed	Ambiguous	Consistency Check
Responsible for	In particular	Unverifiable with the customer policy	Testability Check
Will	Supported by	Unrealistic	
Should (weakest)		Suspected incompleteness	

Figure 2: Incongruous Requirements Checklist (IRC)

In summary, the BA team have found 12 incongruities out of the 23 requirements listed in the PS that need rework. Our Account Manager, Beatrice, had then set up appointments/meetings with specific stakeholders from SUW to clarify the incongruities found. The incongruities are listed in the "Issues" tab of the repository, along with the IRC categories for ease of tracking and reference. This ensures that all identified gaps are documented, visible to the team, and ready to be addressed in subsequent stages.

2.2. Clarify

2.2.1 Role of the Account Manager

The Account Manager (AM) acted as the primary relationship anchor between the vendor team and the customer throughout the engagement lifecycle from the Clarification stage, facilitating communication and supporting effective collaboration.

- **Facilitation of Clarification Sessions (Completed)**
  The AM organised and facilitated structured clarification sessions in collaboration with the vendor team, using pre-agreed agendas, supporting artefacts (e.g., wireframes), and time-boxed discussions. Together with the vendor team, the AM ensured that the appropriate stakeholders, including domain experts and authorised decision-makers, were present to resolve requirement ambiguities efficiently.

- **Expectation Management and Escalation:**
  When certain items, such as baseline data access or security policies, could not be resolved immediately, the AM framed bounded dependencies, managed stakeholder expectations, and escalated issues where necessary.

- **Continuity Safeguard**
  All clarification outcomes were documented in meeting notes and shared repositories. The AM maintains continuity across project phases by preserving contextual knowledge, particularly if team members or stakeholders change roles before testing or deployment.

- **Ongoing Relationship Management**
  The AM maintains regular communication with the customer to understand evolving needs and ensure that the proposed solution continues to align with business objectives.

2.2.2 Stakeholder Engagement

Two face-to-face clarification sessions were conducted with SUW stakeholders (Academic Affairs representative and IT Administrator) on 2nd and 6th February 2026 at SUW. A pre-agreed agenda was circulated 48 hours in advance of each session to ensure discussions remained focused on priority incongruities identified during the breakdown stage.

The first session (2nd February 2026) focused on raising and resolving specific issues identified during the breakdown analysis. These included ambiguities, incomplete requirements, and inconsistencies that could not be resolved through document analysis alone. Such incongruities were assessed as high-risk due to their potential impact on operational correctness and testability.

During this session, paper "napkin" models were sketched to visualise process flows and initial design concepts, including what the scheduling application interface might look like. These informal artefacts facilitated stakeholder discussion by externalising assumptions and uncovering implicit business rules not explicitly stated in the Project Specification.

Following the first session, refined artefacts, including wireframe models, business process models, the improved Data Dictionary, and the ER diagram, were developed to formalise the clarified requirements. These models were used in the second session (6th February 2026), which focused on validating and confirming the updated drafts, evolved ideas, and modelling artefacts with the stakeholders. This iterative validation ensured alignment between stakeholder expectations and the emerging system specification.

At the conclusion of the two sessions, 11 out of the 13 identified issues were resolved. The remaining two issues were documented as open items to be addressed during the subsequent Software Requirements Specification (SRS) stage. [Link to Issues]

2.2.3. Incongruity Outcomes

We have arranged the incongruities identified during the Breakdown stage, together with the actions taken to resolve them in the repository. Each item is documented with its resolution status and a summary of the clarification outcome. This provides a transparent record of which requirements have been fully clarified and which remain open, ensuring traceability and readiness for the Specifying phase.

2.3. Interpret

The vendor team went through the Interpret stage to revise the list of incongruent requirements identified during the Breakdown stage. The result of this activity is to transform the intent into requirements that are consistent and testable for the subsequent codification in the Software Requirements Specification (SRS). Each incongruent requirement was reviewed to determine the scope and governing conditions, while preserving the customer's original intent.

Several incongruencies required a structured transformation. Compound requirements, such as "Record Attendance" and "Automatic Scheduling", were conceptually decomposed into atomic behaviours with clearer actor responsibilities and operational boundaries. Requirements that were ambiguous were interpreted by tightening the scope based on stakeholder clarification, thereby reducing any unwanted assumptions and interpretations. For unseemly prioritisation, we ensured consistency by wording the requirement to match its assigned priority.

For incongruities where clarification remained incomplete or was contingent on unresolved organisational policy, the vendor team avoided making assumptions. These requirements were interpreted only to the extent necessary to isolate and document the outstanding governing conditions. The requirements are left open to be resolved during the Specifying method. As a result, the repository clearly shows which incongruities have been resolved and can be included in the SRS, and which issues are still open that need to be addressed. The revised requirements can be found in the repository.

Revision to Codification standards

During clarification, the customer had agreed with our recommendation to revise the requirements' codification standard and description as follows:

Flagged Requirement (Before)	Revised Requirement (After)
BRA - L1.1: Record Attendance	BO - 1.4.1 Record Attendance
The teaching staff shall be able to capture and record the student attendance automatically using a QR code during the lesson.	The system shall capture and record student attendance automatically using a QR code during lessons, with a fallback option to record manually.

Figure 3: Codification & Description revision format

'BO' represents Business Operations requirements while "IT" represents Technical requirements. The numbering sequence refers to their level of granularity (i.e. Level 0 - 1.x, Level 1 - 1.x.x, Level 2 - 1.x.x.x).

2.4. Categorise

The vendor team set up a structured template in the repository [Link to Github Repo] to organise business operations and technical requirements. All requirements, along with supporting models such as diagrams and process sketches, were fitted into this template. By integrating the models directly with the requirements, the team avoided fragmentation and ensured that both textual and visual specifications were aligned within the same structure.

Incongruities were tracked transparently through Issues, marked open or closed, and the Breakdown, Clarify, and Interpret stages were documented. This collaborative review confirmed completeness and consistency, preparing the requirement set for the Specifying stage.

3. Repository Overview

A repository has been established to support the regulated revision of requirements and ensure safekeeping until the final specification stage. It is organised into dedicated folders for requirements, models, references, and the BCIC analysis stages. This structure provides clarity, transparency, and accountability throughout the development process.

3.1. Structure

- **/Requirement (From Customer PS)** - Contains elicited requirements and their codified versions, organised into subfolders for Software (Registration & Staff Onboarding, Course Administration, Timetable Scheduling, Attendance Management, Monitoring & Reporting), Hardware, Assumptions, and Constraints.
- **/Models**- Stores modelling artefacts, separated into customer_provided (DFD, Gantt chart, use case, etc.) and team_developed (wireframes, organisational chart, and other team-created models).
- **/References** - Holds the original SUW ATAM Project Specification and other supporting materials.
- **/BCIC** - Supports requirement analysis through Breakdown, Clarify, Interpret, Categorise, and the Incongruous Requirements Checklist (IRC).
- **README.md** - Provides navigation and explanation of the repository structure.

3.2. Features

- **Version Control** - GitHub ensures all changes to requirements, models, and analysis documents are tracked safely.
- **Issues & Projects** - Tasks and incongruities are assigned and monitored transparently, supporting accountability.
- **Sanity Check Step** - Acts as a central review point before transitioning to the Specifying phase.
- **IRC Tool** - Provides a systematic way to classify and resolve incongruous requirements.

3.3. Current Status

The repository is now complete and actively in use. All BCIC stages (Breakdown, Categorise, Clarify, Interpret) have been carried through to completion. Many issues raised during development have been closed, with only a very small number left open. Requirements have been codified, models are organised, and supporting documents are properly structured across the repository. Overall, the

repository is stable, comprehensive, and provides all relevant information in a clear and accessible format, ensuring readiness for the Specifying phase and final delivery

4. Risk Analysis

The current status of the project presents several risks that may affect delivery, scope, and stakeholder satisfaction. The Account Manager identified the following customer-side (Singapore University of Waterloo) and vendor-side (CTRL ALT ELITE) risks during engagements.

4.1. Customer-Side Risk

Risk 1: Misalignment due to limited vendor understanding of scheduling operations and constraints.

During early engagements, the risk arises from the vendor not fully understanding the customer's operational workflows, policies (e.g., shift rules, fairness constraints, overtime), and industry-specific practices. This can lead to requirements that don't match real scheduling needs, causing rework, delays, or lower user adoption.

Impact: Medium–High (incorrect requirements, rework, delays, user dissatisfaction)
Likelihood: Medium

Mitigation:

- Conduct structured stakeholder interviews and workflow walkthroughs with schedulers and department heads.
- Provide existing rostering rules, sample schedules, and policy documents early (including edge cases).
- Validate requirements using scenarios and prototypes (e.g., "weekend coverage", "last-minute sick leave replacement").
- Run short requirement review checkpoints after each elicitation session.

Status: Partially mitigated. (Main workflows are understood, but exception rules and policy edge cases are still being clarified.)

4.2. Vendor-Side Risk

Risk 1: Scope creep from frequent requirement changes with the expectation of free-of-charge delivery.

The risk arises from stakeholders continuously adjusting scheduling requirements (e.g., new rule exceptions, new approval flows, additional reporting) and assuming the changes will be delivered within the original scope and cost. This can pressure the vendor team and destabilise the plan.

Impact: High (schedule slippage, cost overrun, reduced quality, team burnout)
Likelihood: Medium–High

Mitigation:

- Establish a clear change control process (impact analysis, approval, and prioritisation).
- Define "baseline requirements" and freeze points per iteration.

- Use a transparent backlog with MoSCoW prioritisation and visible trade-offs. Document assumptions and exclusions clearly in the SOW/contract.

Status: Partially mitigated. (Change process exists, but stakeholders still raise new “urgent” requests.)

Risk 2: Poor requirement quality because customer needs are unclear or hard to articulate.

The customer may be unable to clearly identify, explain, or prioritise scheduling needs at the required level of detail. This can result in vague requirements (e.g., “make it fair” or “optimise staffing”) without measurable rules, leading to misbuilt features and repeated clarification cycles.

Impact: Medium–High (ambiguity, rework, disputes during acceptance)
Likelihood: Medium

Mitigation:

- Use guided elicitation techniques (examples, decision tables, constraint lists).
- Capture rules in testable formats (acceptance criteria, concrete scenarios).
- Run requirement validation workshops with real sample data and schedules.
- Identify a single product owner for fast decisions and prioritisation.

Status: Partially mitigated. (High-level goals are clear, but some constraints and priorities are still not fully defined.)

5. Sign-Off Section

(Authenticator)
Project Lead Signature: _____
Name: _____
Title: _____
Date: _____

Designated Approvers (Name & Signature):

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Yap Hong Jie, Belford _____ Business analyst (External Systems)

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